**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**Cambridge Ordinary Level**

# MARK SCHEME for the October/November 2014 series

## 5070 CHEMISTRY

**5070/22** Paper 22 (Theory), maximum raw mark 75

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2014 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is the registered trademark of Cambridge International Examinations.

CAMBRIDGE
International Examinations

**A1 (a) (i)** S / sulfur / P / phosphorus (1) [1]

**(ii)** Fe / iron (1) [1]

**(iii)** P / phosphorus (1) [1]

**(iv)** Zn / zinc / As / arsenic (1) [1]

**(v)** Fe / iron (1) [1]

**(vi)** H / hydrogen / $H_2$ / N / nitrogen / $N_2$ (1) [1]

**(b) (i)** $4As + 3O_2 \rightarrow 2As_2O_3$ (1) [1]

**(ii)** (arsenous acid) has a lower concentration of hydrogen ions / hydrochloric acid has higher concentration of hydrogen ions (1)

less frequent collisions (between ions in arsenous acid) / more frequent collisions (between ions) in hydrochloric acid (1) [2]

**[Total: 9]**

© Cambridge International Examinations 2014

**A2 (a) (i)** (density generally) increases down the group (1) [1]

**(ii)** allow between 710 – 860 (°C) (1)
(actual value = 760 °C) [1]

**(iii)** liquid (no mark on its own)
melting point is below 35 (°C) **AND** boiling point is above 35 (°C) (1) [1]

**(b) (i)** more reactive down the group / less reactive up the group (1) [1]

**(ii)** $2Rb + 2H_2O \rightarrow 2RbOH + H_2$ (1) [1]

**(iii)** reaction which releases heat / releases energy / products have lower energy than reactants / reaction in which $\Delta H$ is negative / temperature (of surroundings) increases (1) [1]

**(c)** $H^- + H_2O \rightarrow OH^- + H_2$ (1) [1]

**(d) (i)** sodium has low density / nickel has high density (1)

sodium has low melting point / nickel has high melting point / sodium has low boiling point / nickel has high boiling point (1) [2]

**(ii)** any suitable use e.g. manufacture of margarine / other stated hydrogenation reactions e.g. cyclohexane from benzene / sorbitol from glucose / amines from nitro-compounds / amines from nitriles / alkanes from alkenes / alkanes from alkynes (1) [1]

**(iii)** nickel ions are different size to copper ions (1)

idea of disruption of layers in metallic structure / layers cannot slide as easily (1)

**NOTE:** there MUST be some idea of layers / rows or sheets sliding not just atoms sliding [2]

**[Total: 12]**

© Cambridge International Examinations 2014

**A3 (a)** water and salts have different boiling points (1)

water evaporates **AND** salts / residues / impurities / solids left in flask (1)

water condenses / turns to liquid in the condenser (1) [3]

**(b) (i)** $Mg^{2+}$ and $Cl^-$ (1)
**IGNORE:** state symbols [1]

**(ii)** 0.0265 / 0.027 / 0.03 ($mol / dm^3$) (1) [1]

**(iii)** white precipitate / white solid formed / white deposit formed (1) [1]

**(c)** 96 g $SO_4^{2-}$ → 233 g $BaSO_4$ (1)

1.24 g $SO_4^{2-}$ → $\frac{233}{96} \times 1.24$ OR 3.0096 / 3.01 g $BaSO_4$ (1)

mass in 50 $cm^3$ = $3.01 \times \frac{50.0}{1000}$ = 0.151 g (1)

**OR** (for 1st two steps)

moles $SO_4^{2-}$ = $\frac{1.24}{96}$ OR 0.0129 (1)

mass of $BaSO_4$ = $0.0129 \times 233$ OR 3.01 g (1)

**OR**

mass of $SO_4^{2-}$ in 50 $cm^3$ = $1.24 \times \frac{50}{1000}$ OR 0.062 g (1)

moles $SO_4^{2-}$ = $\frac{0.062}{96}$ OR 0.000645833 mol (1)

mass $BaSO_4$ = $0.000646 \times 233$ = 0.151 g (1) [3]

**[Total: 9]**

© Cambridge International Examinations 2014

**A4 (a)** $H^+ + OH^- \rightarrow H_2O$ (1) [1]

**(b) (i)** 20 ($cm^3$) / 0.02 $dm^3$ (1) [1]

**(ii)** mol KOH = $0.15 \times \frac{45}{1000}$ OR $6.75 \times 10^{-3}$ / 0.00675 (1)

mol $H_2SO_4$ = 0.003375 / 0.0034 (1)

concentration = $0.003375 \times \frac{1000}{20}$ = 0.17 / 0.169 (1) [3]

**(c) (i)** ethanoic acid has 1 mol of ionisable H per mol of acid / $H_2SO_4$ has 2 per mol of acid / ethanoic acid is monobasic / $H_2SO_4$ is dibasic / ethanoic acid has one acidic hydrogen (ion) / sulfuric acid has 2 acidic $H^+$ ions / ethanoic acid has half as much ionisable hydrogen (1) [1]

**(ii)** any value between 3 and 6.9 inclusive (1) [1]

**(d) (i)** **ANY TWO FROM**

- sulfur dioxide / $SO_2$ (1)
- (sulfur dioxide) oxidised further / (sulfur dioxide) reacts further to form sulfur trioxide (1)
- oxidation product reacts with water to form sulfuric acid / $SO_3$ reacts with water to form sulfuric acid (1) [2]

**(ii)** irritates skin / irritates eyes / irritates nose / irritates mouth (1) [1]

**[Total: 10]**

© Cambridge International Examinations 2014

**A5 (a)** sodium
barium
magnesium
nickel
copper (1) [1]

**(b) (i)** voltmeter and two wires either side of voltmeter across the electrodes (1) [1]

**(ii)** iron and silver (1) [1]

**(c) ANY TWO FROM**

- the zinc corrodes instead of the iron / zinc reacts instead of the iron (1)
- zinc is more reactive (than iron) / zinc is more reactive (than steel) / zinc higher in the reactivity series (than steel / iron) OR reverse argument (1)
- the zinc loses electrons in preference to the iron (1)

**IGNORE:** sacrificial protection without qualification [2]

**[Total: 5]**

© Cambridge International Examinations 2014

**B6 (a)** sodium chloride is giant ionic structure / has a continuous structure of ions / ions in lattice (1)

strong (attractive) forces between the ions / lot of energy needed to break ionic bond (1)

chlorine is a (simple) molecule / chlorine has simple covalent structure (1)

chlorine has weak forces between the molecules / small amount of energy required to separate molecules / not much energy needed to break intermolecular forces / chlorine has weak van der Waals' forces (1) [4]

**(b)** in molten sodium chloride ions can move but ions can't move in solid / ions can only move in molten sodium chloride (1) [1]

**(c)** sodium ion 2, 8 and + charge (1)
chloride ion 2, 8, 8 and – charge (1) [2]

**(d)** at the negative electrode / cathode reduction takes place which is gain of electrons (by sodium) (1)

at the positive electrode / anode oxidation takes place which is loss of electrons (by chloride) (1)

**OR**

sodium ions are reduced because they gain electrons (1)

chloride ions are is oxidised because they lose electrons (1)

**OR**

sodium is reduced because oxidation number of sodium decreases (1)

chloride / chlorine is oxidised because the oxidation number of chlorine increases (1) [2]

**(e)** $2NH_3 + 3Cl_2 \rightarrow N_2 + 6HCl$ (1) [1]

**[Total: 10]**

© Cambridge International Examinations 2014

**B7** **(a)** alkenes (1) [1]

**(b)** melting points increase (1)

increase in melting point from even number to odd number of carbon atoms is less than from odd to even number / the increase is less for some atoms than others / any reference to the regular zigzag nature of the increase (1) [2]

**(c)** $C_9H_{20}$ (1) [1]

**(d)** **(i)** $C_{11}H_{24} \rightarrow C_2H_4 + C_3H_6 + C_6H_{14}$ (1) [1]

**(ii)** **ANY TWO FROM**

- (hydrocarbons with) longer chains not in high demand / more longer chains produced than used / shorter chains in more demand / fewer short chains produced than used (1)
- so (more) petrol / gasoline is made (1)
- to produce alkenes / to make ethane (1) [2]

**(e)** **(i)** 16 g methane $\rightarrow$ 27 g HCN (1)

500 g methane $\rightarrow 500 \times \frac{27}{16} \times \frac{65}{100} = 548$ g (1)

**OR**

$\frac{500}{16} = 31.25$ mol methane (1)

$31.25 \times 27 \times \frac{65}{100} = 548$ g (1) [2]

**(ii)** $Ca(OH)_2 + 2HCN \rightarrow Ca(CN)_2 + 2H_2O$ (1) [1]

**[Total: 10]**

© Cambridge International Examinations 2014

**B8 (a) (i)** concentration of ethanoate = 0.45 mol / $dm^3$ (1)

mass = $0.45 \times 59 \times \frac{200}{1000}$ = 5.31 / 5.3 g (1) [2]

**(ii)** $\frac{0.17}{300}$ = $5.67 \times 10^{-4}$ / $5.7 \times 10^{-4}$ (mol / $dm^3$ / s) (1) [1]

**(iii)** rate of reaction decreases with time / reaction slows down (1)
concentration (of $H^+$ ions) decreases / concentration (of reactants) decreases / concentration (of ethyl ethanoate) decreases (1)
collision frequency reduced (1) [3]

**(b)** $Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$
correct formulae (1)
correct state symbols (dependent on correct formulae) (1) [2]

**(c)** filter (off iron) (1)

heat filtrate to crystallisation point then leave to crystallise / evaporate off some of the water from filtrate then leave to crystallise / partially evaporate filtrate and leave to crystallise
**AND**
dry crystals with filter paper (1) [2]

**[Total: 10]**

© Cambridge International Examinations 2014

**B9** **(a)** decreases with increase in temperature (1)

reaction is exothermic / increasing temperature favours reaction which absorbs heat (1) [2]

**(b)** increases with increasing pressure (1)

increasing pressure causes reaction to go in direction of decreasing number of moles / smaller volume (1) [2]

**(c)** **ANY ONE FROM**

- low(er) temperature makes reaction rate too slow (1)
- high(er) temperature decreases percentage yield (1)
- low(er) temperature increases percentage yield (1)
- this temperature (i.e. 350–450) gives a (relatively) high rate and low yield (1)

**ANY ONE FROM**

- low(er) pressure gives poor yield (1)
- high(er) pressure increases yield (1)
- high(er) pressure expends too much energy (1)
- high a pressure too expensive (1)
- high(er) pressure gives a higher rate (1)
- high pressure a safety risk (1)
- this pressure (i.e. 200–300) gives a high yield and high rate (1) [2]

**(d)** speeds up the reaction / lowers the activation energy (1)

lowers energy costs / less energy used (1) [2]

**(e)** molar mass of $(NH_4)_3PO_4$ = 149 (1)

$\frac{42}{149} \times 100 = 28.19\% / 28.2\%$ (1) [2]

**[Total: 10]**

© Cambridge International Examinations 2014